CONFIGURING FORTIAP USING FORTIGATE

Xiao, Cambell
Cisco CCNP
period 3-4

Purpose

The purpose of this lab is to configure and deploy 2 SSIDs, one running on WPA2 and one running on Enterprise in a school environment where our filters are applied to our SSIDs. One SSID should run on a RADIUS server and one using a pre-shared key.

Background Information

A wireless access point is usually connected to the upstream network to perform (as its name implies) as an access point to get in. Usually, APs will broadcast wireless networks for hosts to connect to and send their data through the access point upstream through beacons. While these APs are broadcasting these beacons, the network will appear as WIFI for end users to connect to. The user can authenticate themselves into the network in multiple ways however in this environment, we only used two: RADIUS and WPA2. WPA2 uses a pre-shared to authenticate into the network. The user will enter a configured key configured on the AP to connect to the network and all network traffic sent towards the AP will be encrypted so malicious users running programs such as packet tracer can't easily intercept the traffic and immediately see all data sent across. RADIUS works in a much different way, where it is a sort of add-on to WPA2. RADIUS itself does not provide encryption in between the AP and the host, but its purpose is to make sure that all users on the network is accounted for and observed, so any users that could possibly not be related to the corporation hosting the network will immediately be identified. A RADIUS server will only allow certain users onto the network, meaning hosts will have to input a username and a password to access the network. It will then authorize the user a certain level of access into the network, which can be preconfigured whilst creating the user. Once the user is logged into the network, the RADIUS server can log any packets the user sends to the internet through the AP. This makes it extremely easy to notice if anyone malicious is connected to the network. If an IP address that is not listed as a part of the corporation is found, then it's simple to identify them as an outside entity, likely a malicious entity. This way, RADIUS provides triple A protection for any SSIDs created with the RADIUS server. However, because RADIUS does not encrypt traffic, it's important to run an encryption method whilst running RADIUS at the same time such as WPA2 or WPA3.

Now that we have established the general premise of how access-points work, its important to understand how FortiGate access-points differ. Usually, access-points are configured by accessing the GUI after a direct ethernet connection. From there, SSIDs can be directly configured. However, Fortinet allows access points to be directly configured through the Fortigate firewall. This provides much more accessibility because when a WAP is connected into the network of a Fortigate firewall, the firewall will immediately detect the WAP and allow manual configuration through the firewall. This means that instead of having to manage each AP individually over webpages, the APs can all be managed through the FortiGate.

Creating authentication services through Fortigates are another strong suite of Fortinet. A local RADIUS server can be configured on the firewall, which further benefits user accessibility because instead of having to manage a remote RADIUS server, all settings can be changed directly on the firewall. A local RADIUS server also allows ease of connectivity between SSIDs and the RADIUS server because the server exists on the Fortigate firewall. This means that if the firewall has connection to the AP, the RADIUS server will stay up. This is particularly beneficial because if the access point cannot contact the

firewall, then the SSIDs will immediately go down. Thus this feature provides 24/7 authentication services for the SSIDs as long as the wireless connections are up.

When access points are connected to an upstream network, the SSIDs that are created on the access-point will often exist on separate networks than the network of upstream internet. An example of this could be a host connected to an access point with a network of 10.0.0.0/24, whilst the upstream network exists on the 192.168.1.0/24 network. By default, these two networks are unable to communicate with each-other because they are in separate subnets. For these two networks to communicate, all access-point data with a source address of 10.0.0.0/24 have to be converted to a 192.168.1.0/24 address, particularly through NAT. Fortigate automatically does this when exiting the network through their policies. Instead of having to manually configure NAT in a layer 3 switch between the Fortigate and the WAP, the Fortigate will automatically function as a layer 3 routing device and NAT the data from the AP to the outgoing interface of the Fortigate. This makes it particularly easy for network admins to configure new SSIDs because every new SSID doesn't need a separate NAT statement to transfer networks.

Lab Summary

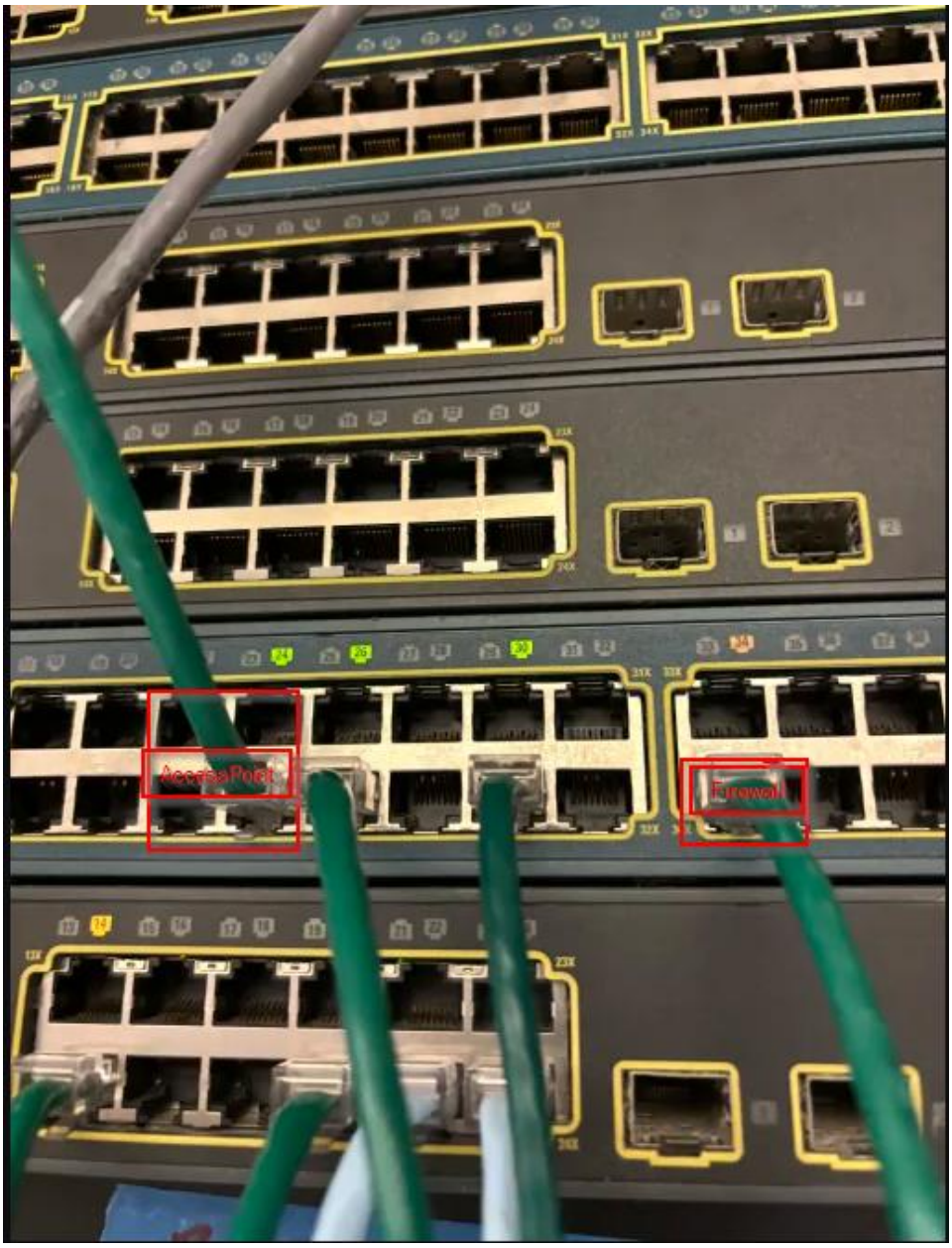
Unlike how many other access points work, FortiAPs can directly be configured through the FortiGate firewall if they have a direct connection. Thus, the first step was to connect the FortiAP to the FortiGate through a switch. Under Wifi & Switch Configurations > Managed FortiAPs, the AP should pop up and have the status of "online". We entered Wifi & switch controller > FortiAP Profiles and created a profile. We entered the serial number of the AP then set it to use tunnel mode with 2.4G and 5G. Then we entered Wi-Fi & Switch controller and created a new SSID, inputting a name, a subnet, and a passphrase. Then the SSID should be up and broadcasting. To get internet on the SSID, we created a firewall policy that pointed from the SSID to the internet and the internet to SSID, allowing all processes. Auto-NAT should be configured on the outgoing interface to get past the difference between subnets.

Lab Process

Lab Commands

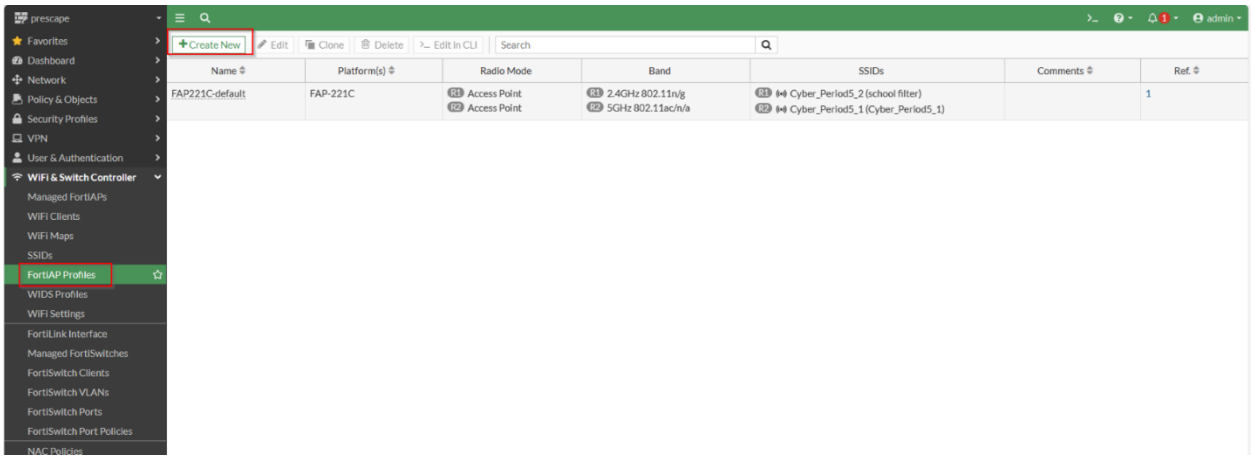
1.

Connect the firewall and the AP with a direct connection through a power over ethernet switch.



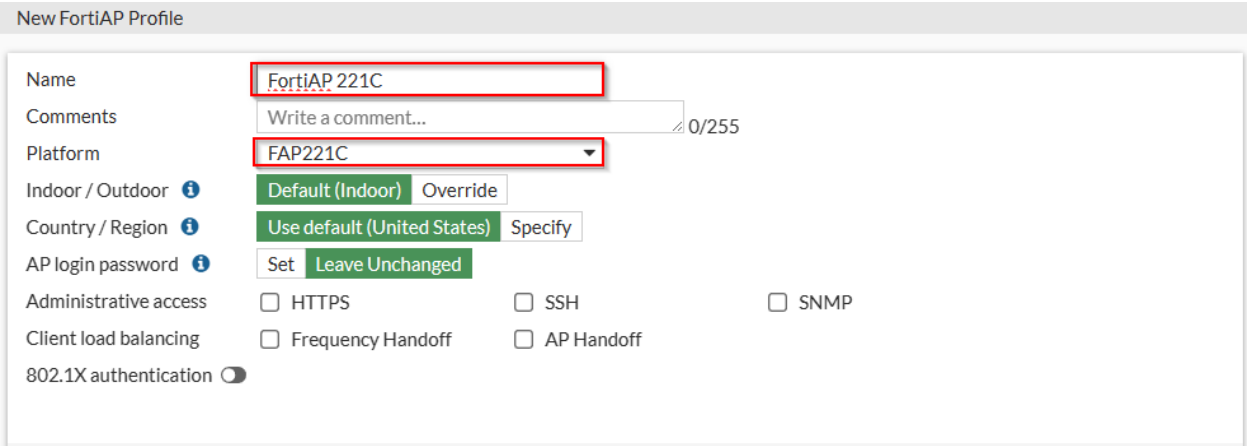
2.

Under WIFI & Switch controller > FortiAP Profiles, select “Create New”



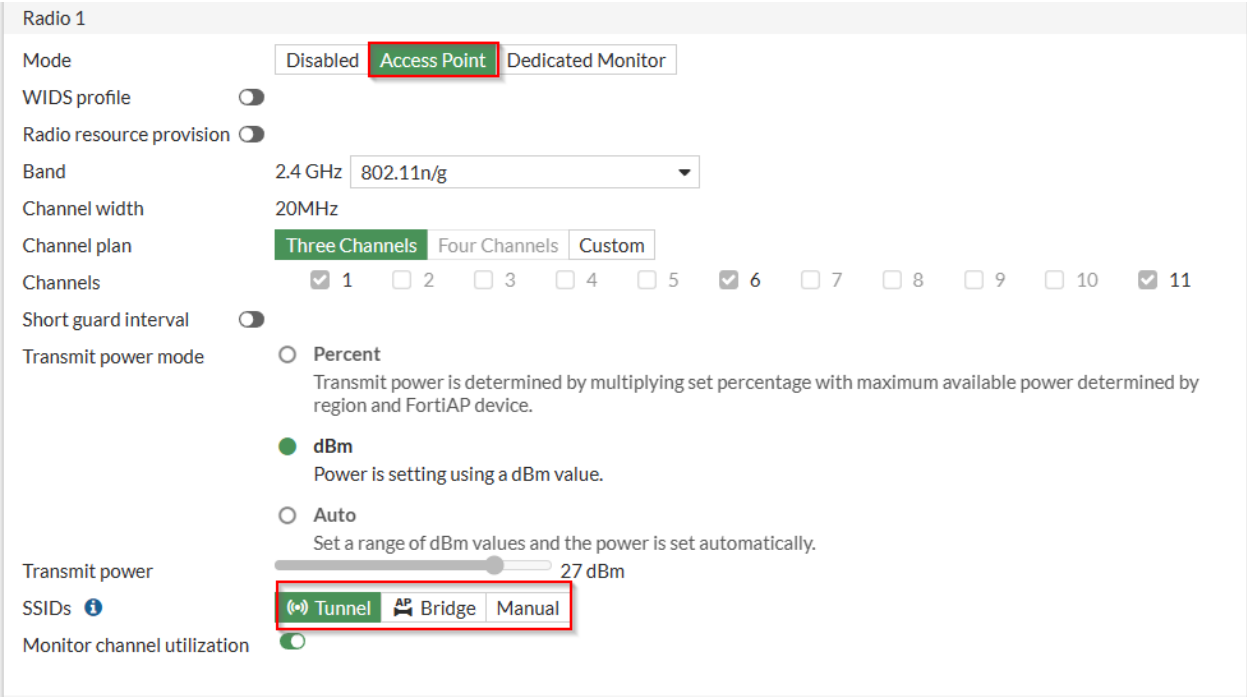
3.

Create a name for the FortiAP and select the platform that it runs on.



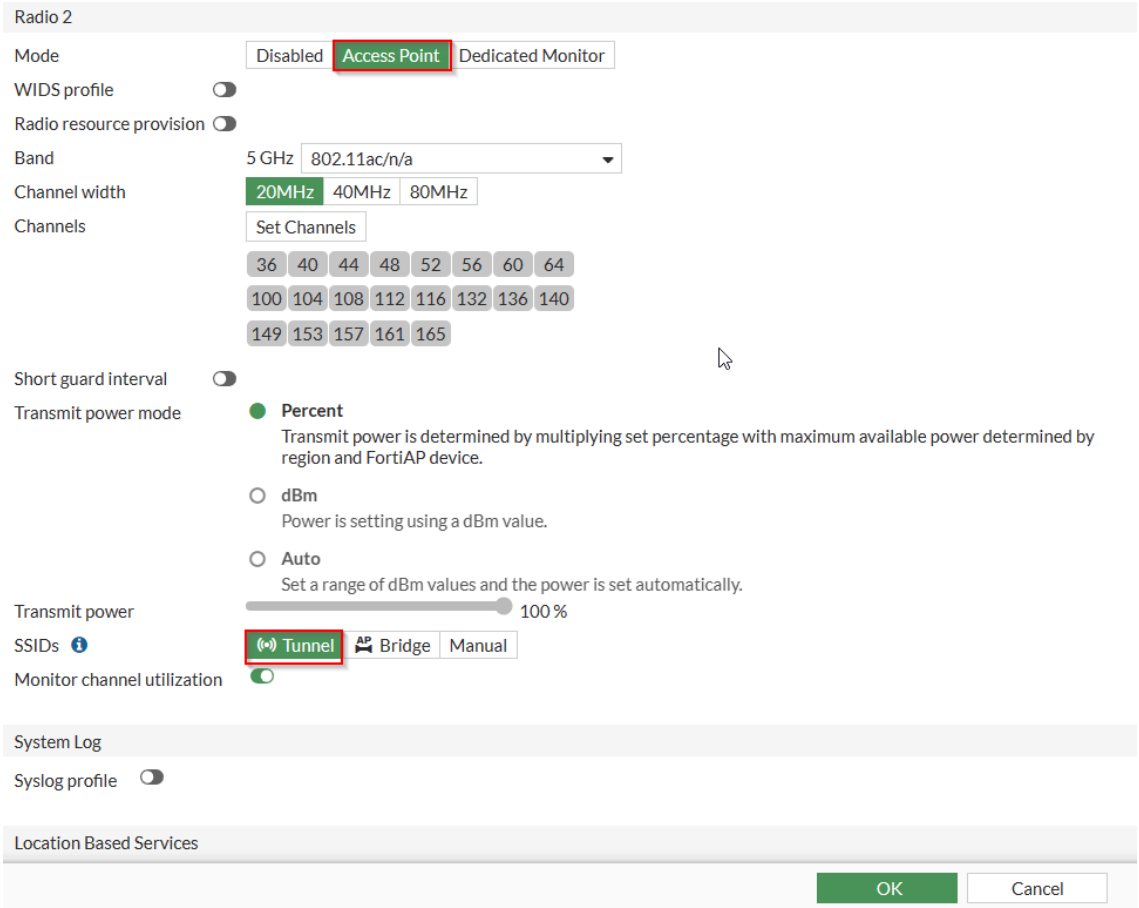
4.

Under Radio 1 which uses 2.4GHz, set the mode to Access Point and the SSID mode to Tunnel mode. This will allow broadcasting of 2.4 GHz SSIDs.



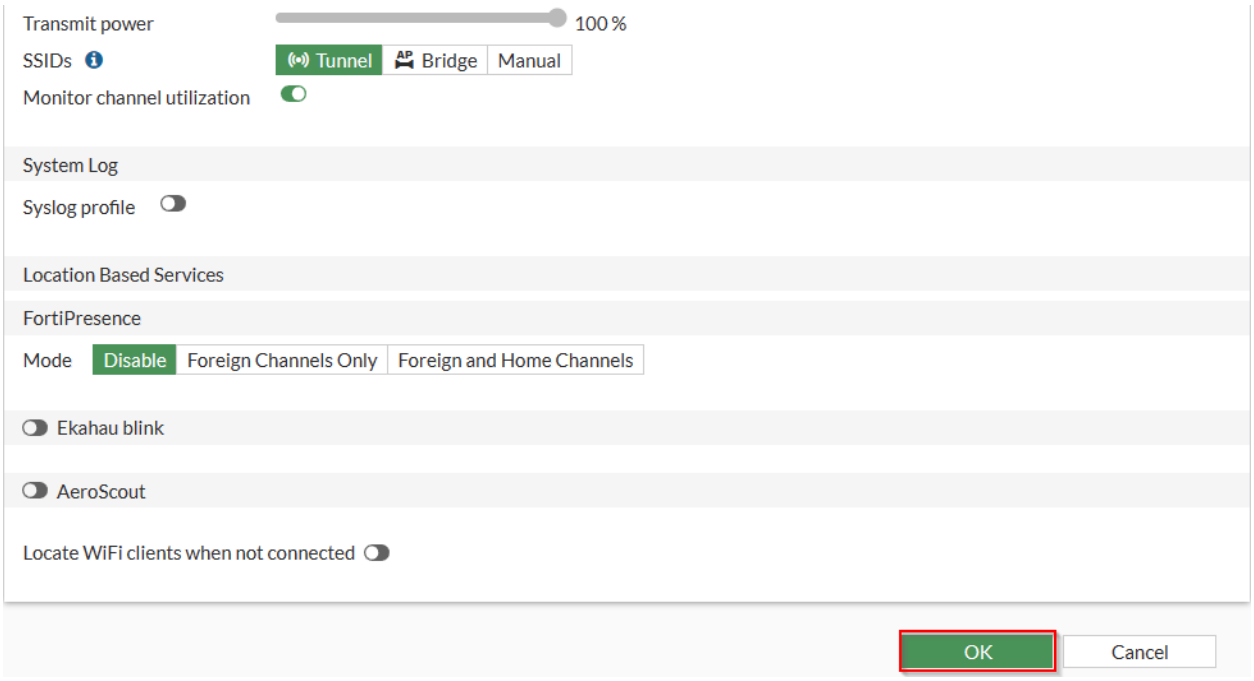
5.

Under Radio 2, which runs 5GHz, set the mode to Access Point and then set the SSID to Tunnel mode. This will allow the broadcasting of 5GHz SSIDs.



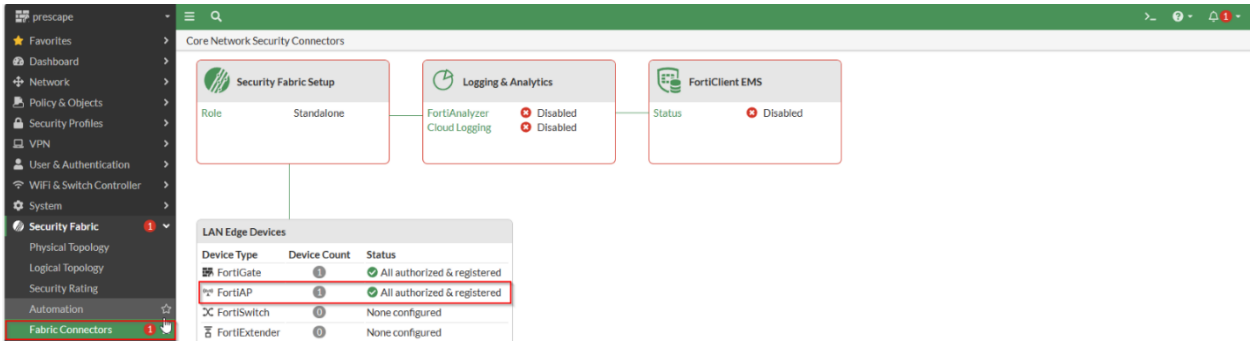
6.

Press OK to finish the configuration process for the AP.



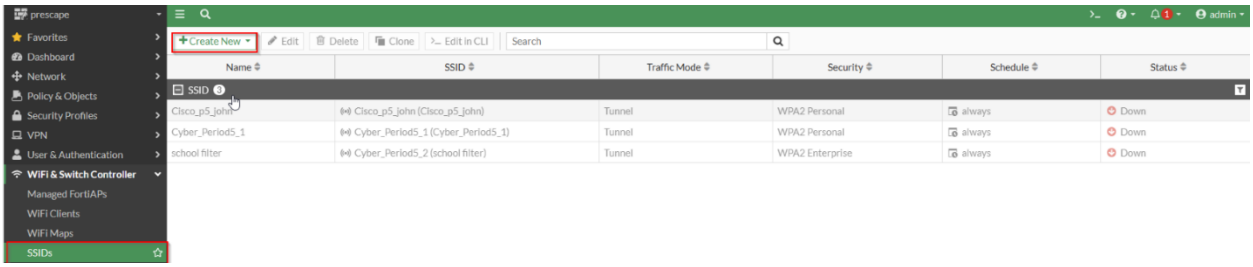
7.

Under Security Fabric > Fabric Connectors, verify that the FortiAP is now authorized and registered.



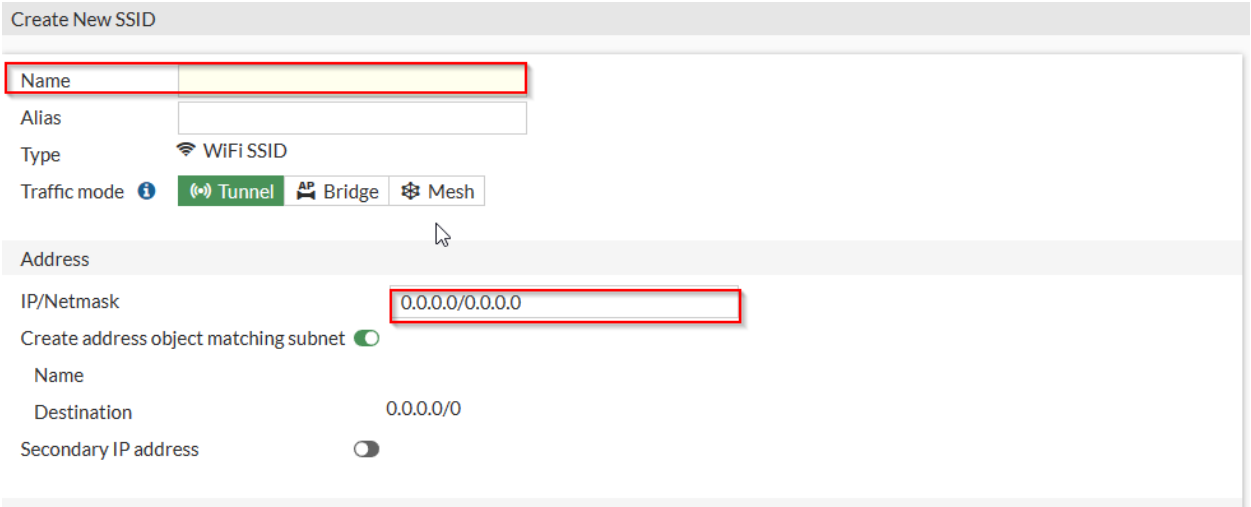
8.

Under WIFI and Switch Controller > SSIDs, select “Create New” to create a new SSID.



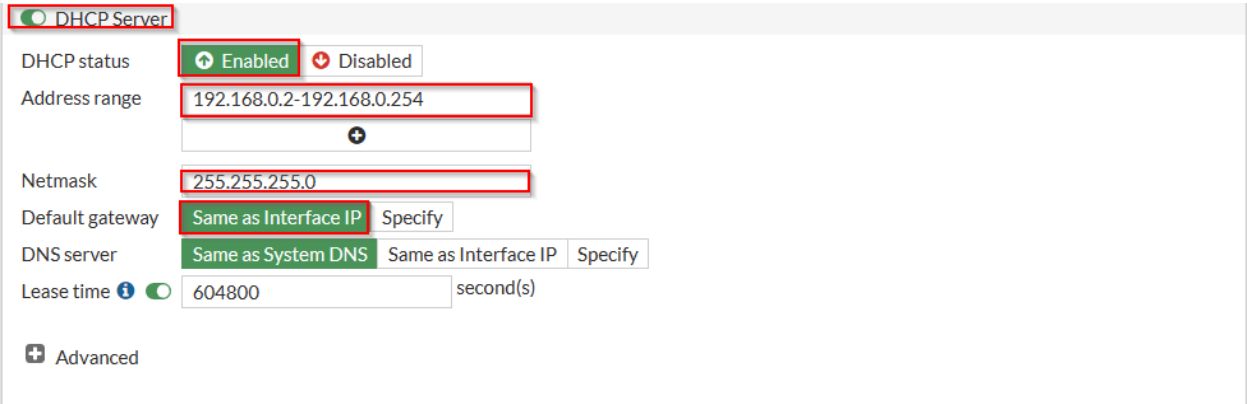
9.

Create a Name and the DHCP server address



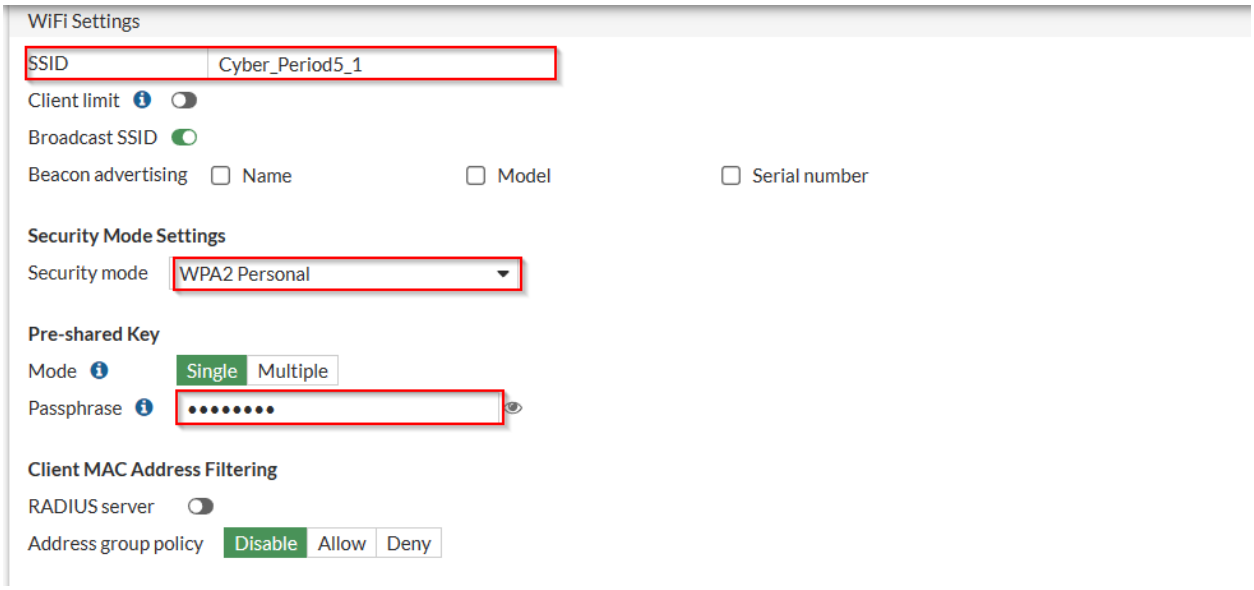
10.

Enable DHCP Server and enable DHCP. Select the Address Range at which hosts in the network will run on and the netmask.



11.

Under WIFI Settings, set an SSID, the security mode, and the preshared key.



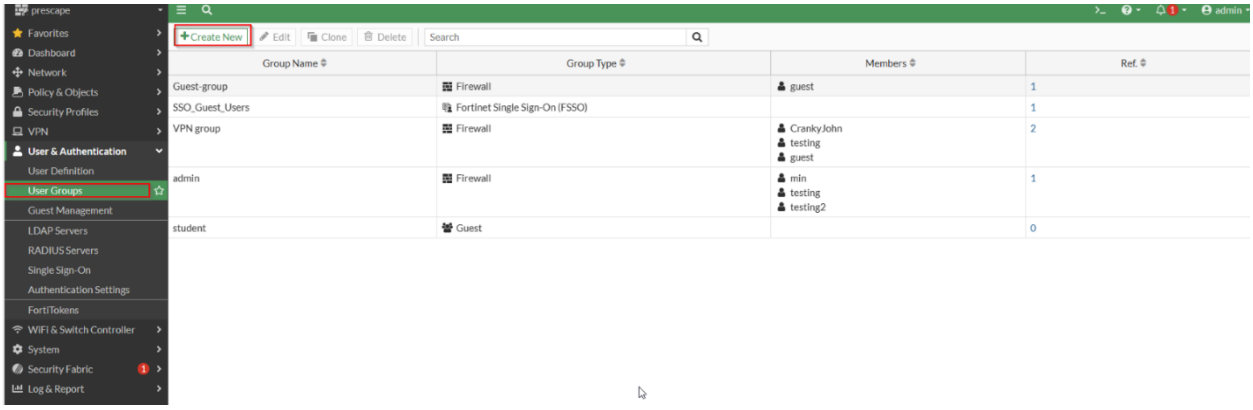
12.

press OK to finish configuring this SSID.

Creating RADIUS SSID

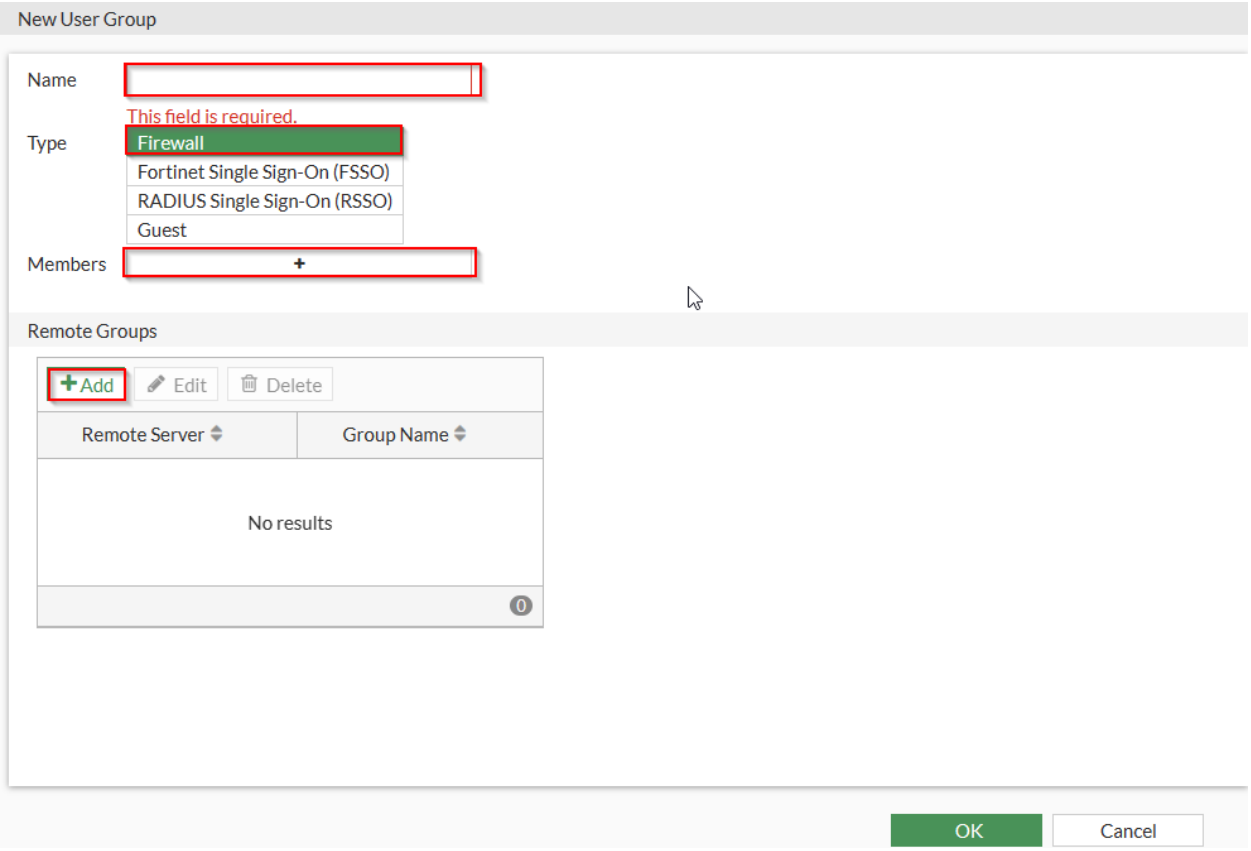
13.

Enter User & Authentication and create a new User group.



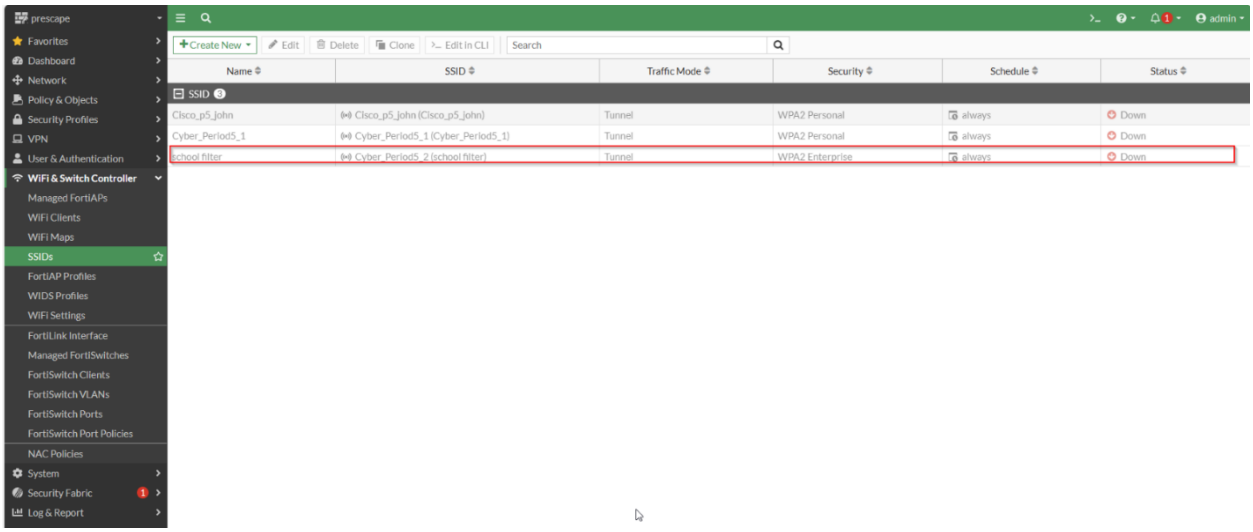
14.

Create a name and set the usergroup type to firewall and add any members. Remote groups will not be necessary for our lab but adding them is optional.



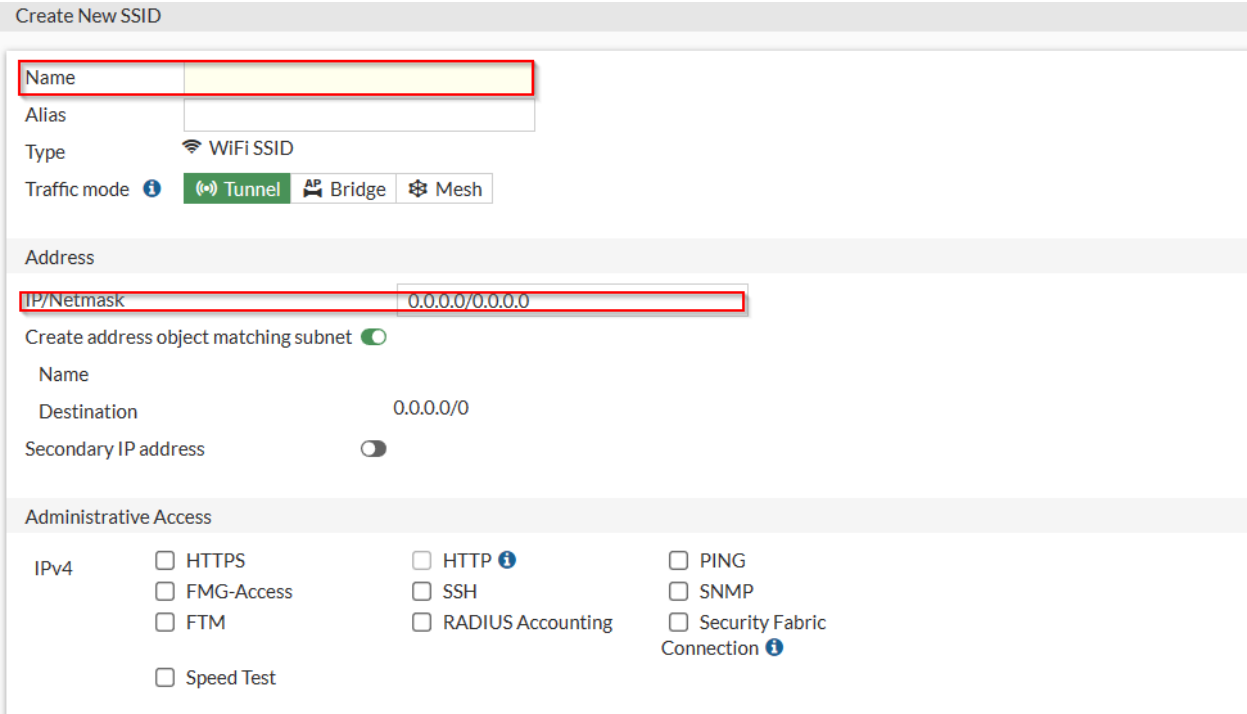
15.

Return to Wifi & Switch Controller and add a new SSID.



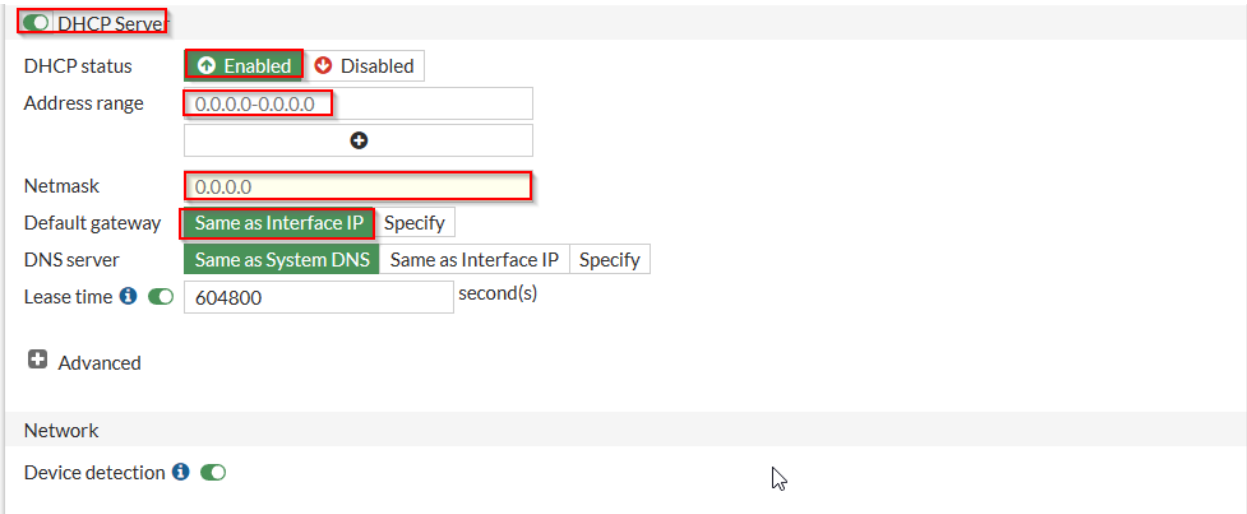
16.

Set a name for the SSID that will appear in fortigate. Then set the ip address of the default gateway of the network.



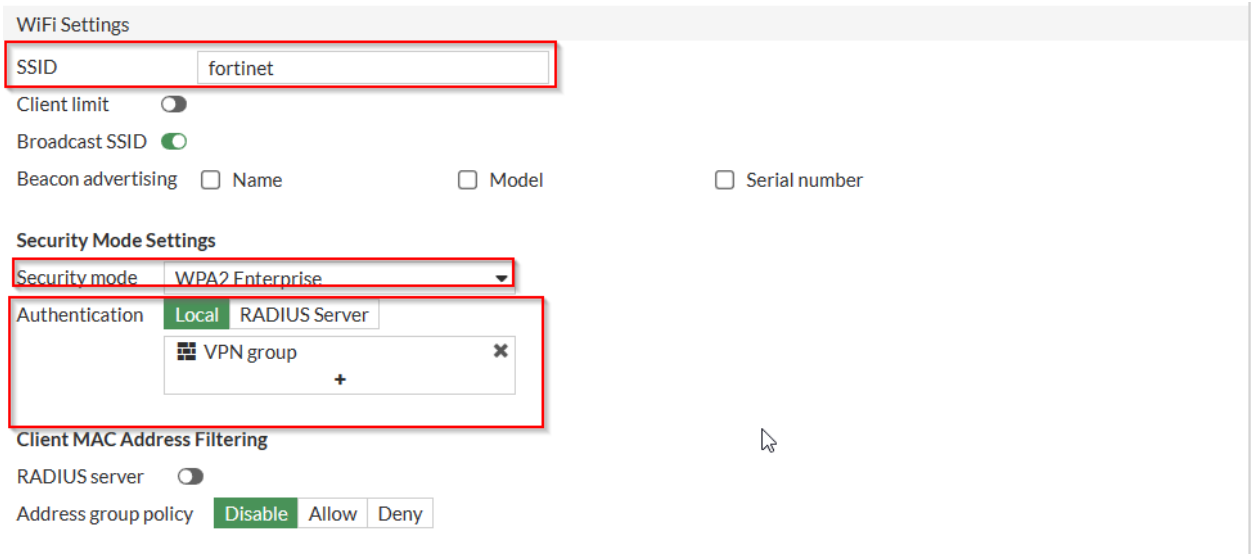
17.

Enable the DHCP Server and set the address range which devices on the network will run on. Then set the Netmask



18.

Create the SSID broadcast name and set the security mode to WPA2 Enterprise. Set the authentication mode to local and select which group is allowed to access this SSID.

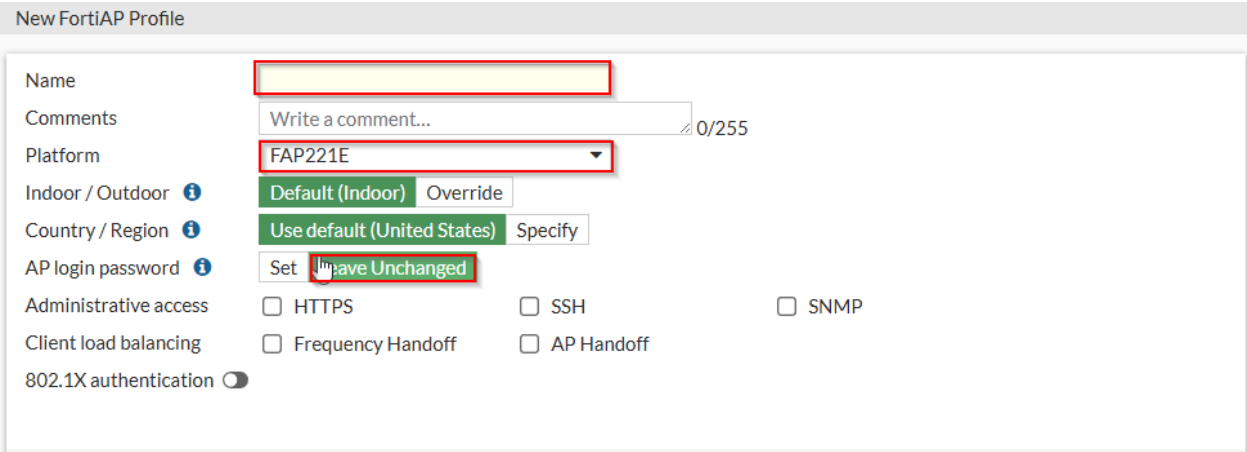


19.

Enter WIFI & Switch Controller > FortiAP Profiles and select “create New”

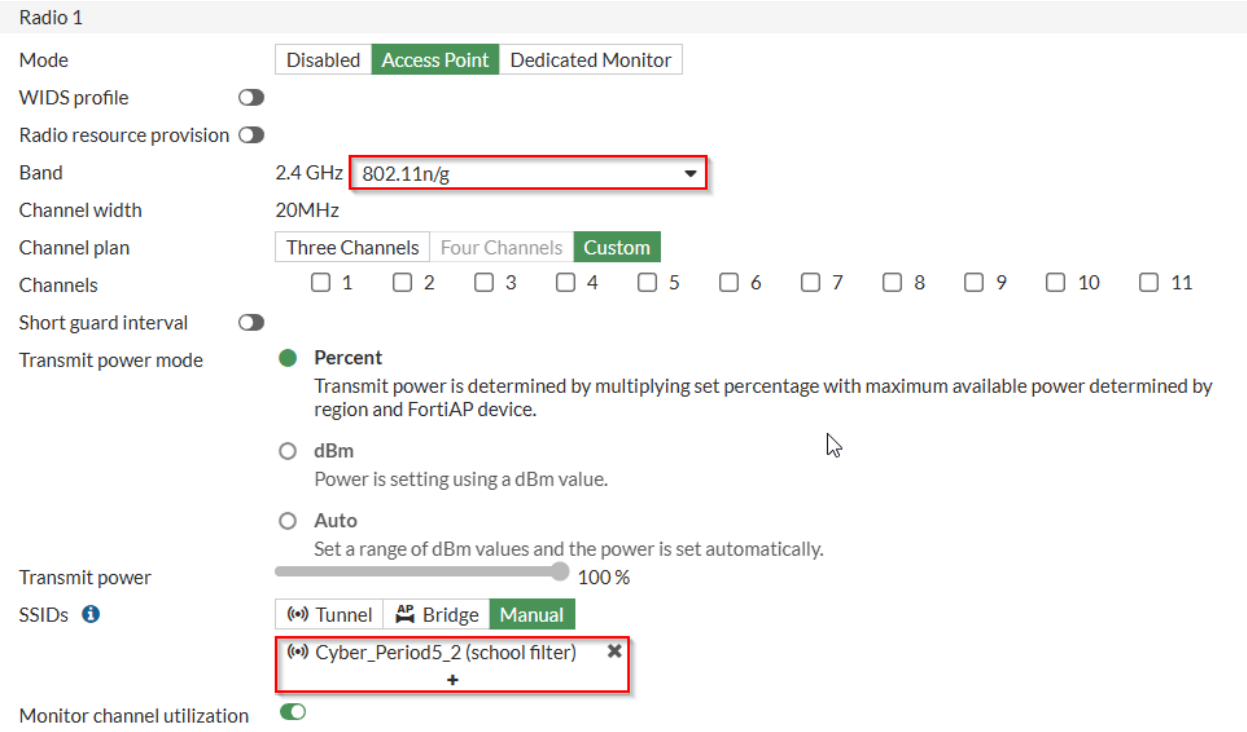
20.

Enter a FortAP name and the platform of the AP.



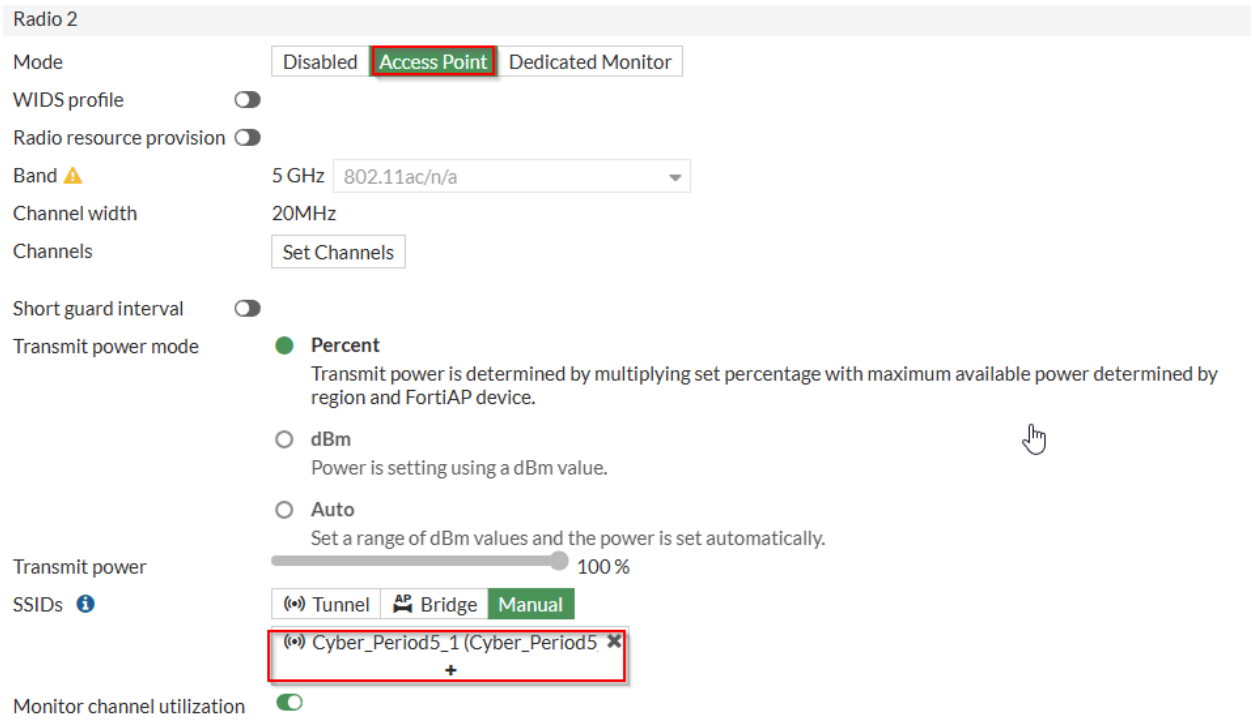
21.

Set the radio mode to Access Point and set the 2.4 GHz band to 802.11n/g. Then set the manual SSID that will run on 2.4GHZ.

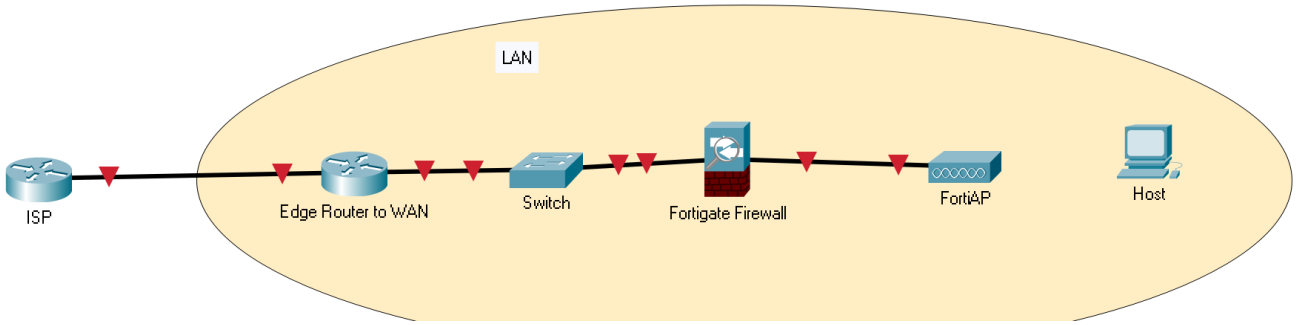


22.

Enter Radio 2 and set the mode to Access Point. Set an SSID to run on 5G. Then select OK.



Network Diagram



Problems

SSIDs require a network policy to get in/out of the network

By default, Fortigate firewalls automatically refuse to forward any traffic from the LAN to the WAN or vice versa. This is used because network administrators are unable to block every malicious connection, thus an implicit deny rule would be very useful. We needed to manually

create two policies: one to enter the SSIDs from the WAN and one to enter the WAN from the SSIDs. This allowed devices on our network to access the internet.

FortiAP takes forever to connect to the FortiGate.

When we first connected the FortiGate and the FortiAP together, we underestimated the amount of time it took for these to connect so we thought we did something wrong. Turns out after waiting for around 30 minutes for the access point to connect, the firewall finally received the connection and registered it.

Conclusion

In comparison to majority of other wireless access points, Fortinet's ease of access is an absolute dealbreaker when attempting to expand your network. Not only do Fortigates automatically connect to any access points connected to It, the configuration of SSIDs is incredibly easy. Coupled with the ability of the Fortigate to auto-NAT outside of the network, there are very little steps a network administrator needs to do to get a FortiAP on their network. A Fortigate firewall has an incredibly easy GUI to use, from being able to configure firewall policies, SSIDs, managing any Fortinet devices on the network, and VPNs with a click of a button.